Predicting fine-tuning generalization from activation geometry

Decoding Dan's M-update framing

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The proposal

Represent a *context* as an activation vector v_c (read at the token after the last user-message token) and a *behavior* as an activation vector v_b (read at the token after the assistant-message token). Suppose that, before training, the model's context $\to$ behavior map is approximately linear,

$$v_b' = M\, v_c'.$$

Dan's conjecture: after fine-tuning context c' to produce a new behavior v_b'' (instead of its original v_b'), the map becomes a rank-one correction of the old one,

$$\boxed{M_{\text{new}} \;=\; M \;+\; (v_b'' - v_b')\, v_c'^{\top}}$$

The question is whether something of this shape predicts how training on $c_0 \to b_0$ generalizes to a different $c_1 \to b_1$.

What the equation actually claims

The update is a rank-one outer product: the behavior shift $(v_b'' - v_b')$ times the trained context $v_c'^{\top}$. Evaluate the new map at the training point:

$$M_{\text{new}}\, v_c' = M v_c' + (v_b'' - v_b')\, \langle v_c',\, v_c' \rangle = v_b' + (v_b'' - v_b')\, \|v_c'\|^2.$$

This exactly hits the target v_b'' only when $\|v_c'\| = 1$, so the framing wants to live in *normalized / cosine* space with a fitted scalar gain. Now evaluate at a *new* context c_1:

$$v_b(c_1) = M v_{c_1} + (v_b'' - v_b')\, \langle v_c',\, v_{c_1} \rangle,$$

so the predicted **change in behavior at any new context** is

$$\boxed{\Delta v_b(c_1) \;=\; (v_b'' - v_b') \,\cdot\, \langle v_c',\, v_{c_1} \rangle}$$

Two falsifiable claims fall out:

1. **Direction is constant.** The behavior shifts in the *same direction everywhere* — the direction it was pushed at the training point. Only the magnitude varies across contexts.
2. **Magnitude $\propto$ context cosine.** How much training on c' bleeds into c_1 scales with the similarity $\langle v_c',\, v_{c_1} \rangle$ of their (base-model) context vectors.

The clean bridge to what we already measure

We do not measure v_b directly; we measure $\Delta \log P(\text{marker})$ on-policy at the post-response slot. If the behavior is linearly readable — the persona-vector premise — there is a probe direction w with $\ell(c) = \langle w,\, v_b(c)\rangle$ for the behavior logit. Then

$$\Delta\ell(c_1) = \langle w,\ v_b'' - v_b'\rangle \cdot \langle v_c',\, v_{c_1}\rangle = \underbrace{\langle w,\, \Delta b\rangle}_{\text{one scalar per behavior}} \cdot \langle v_c',\, v_{c_1}\rangle.$$

That is a **one-free-parameter prediction**: leakage to every bystander context should be *linear in its base-model cosine to the trained context, with a single shared slope.* Directly testable against the existing leakage tables (#432→#456, #448): regress measured $\Delta \log P(\text{marker})$ per persona against base-model context cosine; check linearity-through-origin with one slope.

This is the mechanism for #404/#458, not a new direction

The persona-distance predictor line (#404/#458: does base-model KL/JS/cosine predict leakage?) is the *empirical* version of exactly this. Dan's M-update is the generative model that says *why* distance predicts leakage, and it sharpens the existing result three ways:

- predicts the relationship is **linear in cosine**, not merely monotonic;
- predicts a **shared slope** across contexts for a fixed behavior;
- adds genuinely new, untested content — the **directional claim**: the activation-space shift is the *same direction* everywhere. Current predictors only touch magnitude. Whether $\Delta v_b(c_i)$ is rank-one / constant-direction is the part nobody has checked.

It also explains contrastive negatives (#18/#207/#383)

Positive-only training gives $M + \Delta b\, c_{\text{src}}^\top$. Leakage at bystander c_j is $\Delta b \cdot \langle c_{\text{src}},\, c_j\rangle$; since persona contexts share a high baseline cosine, you get *uniform* leakage — exactly the #18/#207 observation (92–98% on all bystanders). Contrastive negatives add **negative rank-one terms** at the bystander contexts that cancel the $\langle c_{\text{src}},\, c_j\rangle$ term, which is why the distance→leakage gradient exists *only* in the contrastive regime. The framing predicts this for free:

$$M_{\text{new}} \approx M + \Delta b\, c_{\text{src}}^\top - \sum_j \varepsilon_j\, c_{\text{byst},j}^\top.$$

Where it is "a bit wrong" (Dan's hedge), ranked by impact

1. **Direction rotation is the real risk.** The strong claim is a context-independent shift direction. Plausibly it rotates — evil-on-coding could land as "evil" nearby but "off-target weird" far away. First thing to measure: SVD the matrix of realized $\Delta v_b(c_i)$ and check it is rank-one with right vector $\approx v_c'$ and left vector $\approx \Delta b$.
2. **Rank-one sufficiency.** SGD distributes the update across all weights, so the induced map-change may be higher rank. ROME/MEMIT show narrow edits *are* roughly rank-one, so it is a reasonable prior — but it is the load-bearing assumption.

3. M **itself shifts = emergent misalignment.** The model assumes M is preserved except along v'_c. EM is precisely the failure of that: narrow training broadly rewrites M. In-framing, EM corresponds to either (a) a trained context v'_c with high mean similarity to the whole context distribution (a generic context $\Rightarrow$ broad leakage), or (b) a realized shift with a large off-rank-one component. A concrete, testable account of *when* narrow training generalizes broadly.

4. **Normalization & layer choice.** Needs cosine + fitted gain, and a sweep over which layer / layer-pair gives the cleanest linear M. Context and behavior may read best at different layers.

5. **Lossiness.** A single-token v_c/v_b may not capture the full context/behavior, and emission goes through softmax + sampling. The linear-probe bridge handles the last step, but only if the behavior is genuinely linearly readable.

Minimal experiment to nail it

One pod, one clean test:

- Fix a behavior (the marker is cleanest — already measured on-policy). Train a single source context $c_0 \to$ marker.
- Across N held-out contexts (the persona panel), measure: (a) realized $\Delta \log P(\text{marker})$; (b) realized activation shift $\Delta v_b(c_i)$; (c) *base-model* context cosines $\langle v_{c_0}, v_{c_i} \rangle$.
- **Headline test:** predict held-out leakage using *only* base-model quantities plus the single training pair. If that prediction correlates with measured leakage across held-out contexts, the framing is predictive. Report (i) variance explained by the rank-one / cosine model, (ii) whether the shift direction is constant, (iii) the residual = the off-target / direction-rotation part.

Even a partial result ("magnitude is cosine-predictable, direction rotates by X°") is publishable and tells us something real.

Take

Worth pursuing, as the *deep* one. It (a) unifies three threads under one generative model, (b) upgrades #404/#458 from "distance correlates" to "here is the mechanism + a directional prediction," (c) is Dan's own framing, hence maximally aligned, and (d) is exactly the "fewer experiments, nailed carefully" shape he keeps asking for — feeding the Aug 31 paper-draft milestone. Caveat if it stalls: do not over-invest in the *exact* rank-one form. The valuable deliverable is "to what extent is fine-tuning generalization linearly predictable from base-model activation geometry"; the rank-one update is one hypothesis within that, not the whole prize.